Post-Completion Write-Up (Initial Batch)

Initial Batch Migration Summary

Overview

Between April 20 and May 4, the Kali Linux, Metasploitable 2, and Windows Server 2022 VMs were fully hardened and prepared for Azure migration. This batch forms the core of the ACME Cybersecurity Lab's infrastructure.

Key Hardening Actions

- Kali Linux: SSH key-only access, password hardening, Fail2Ban implementation, system snapshot
- Metasploitable 2: Service kill procedures, UFW firewall configuration, SSH hardening, strong password enforcement
- Windows Server 2022: GPO implementation, enhanced password policies, RDP security hardening, unnecessary service disablement, comprehensive event logging

Migration Method

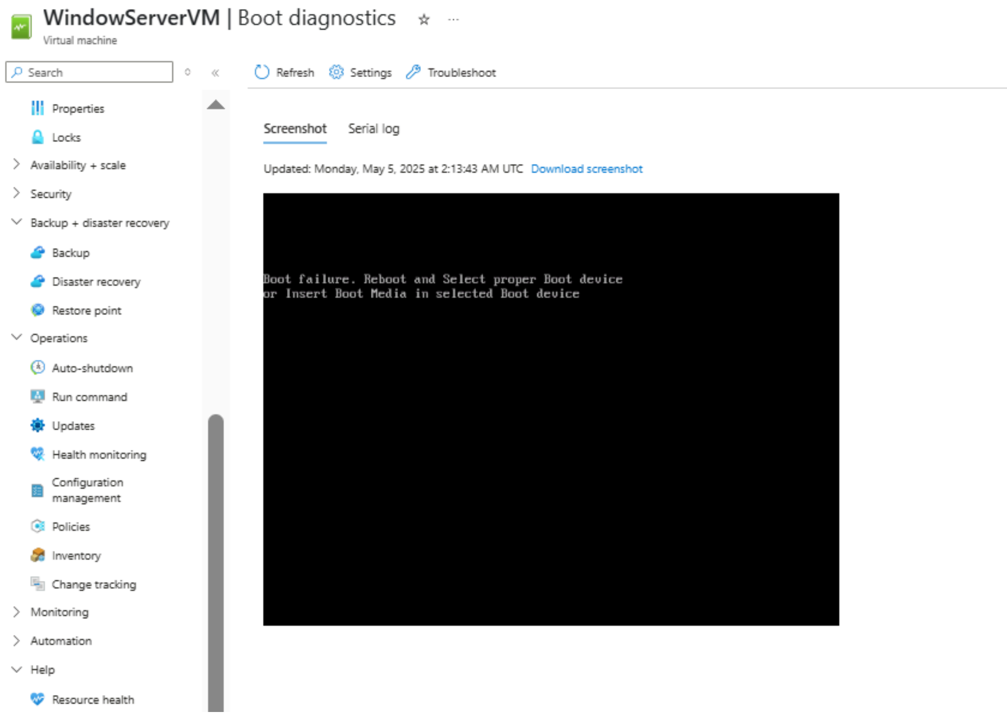
- All VMs were exported from VMware Workstation
- Converted using StarWind V2V Converter to VHD format
- Uploaded via Azure Storage Explorer as managed disks

Technical Challenges and Resolution Attempts

Primary Boot Failure Issue

The most significant challenge was that VMs uploaded as managed disks would not boot after conversion to Azure VMs. Root causes identified:

Conversion Limitations: StarWind V2V Converter lacks image generalization capabilities (sysprep equivalent), which is critical for cloud deployment. The tool doesn't strip hardware-specific drivers or configurations, causing Azure's hypervisor to fail during boot due to driver conflicts and hardware abstraction layer (HAL) mismatches.

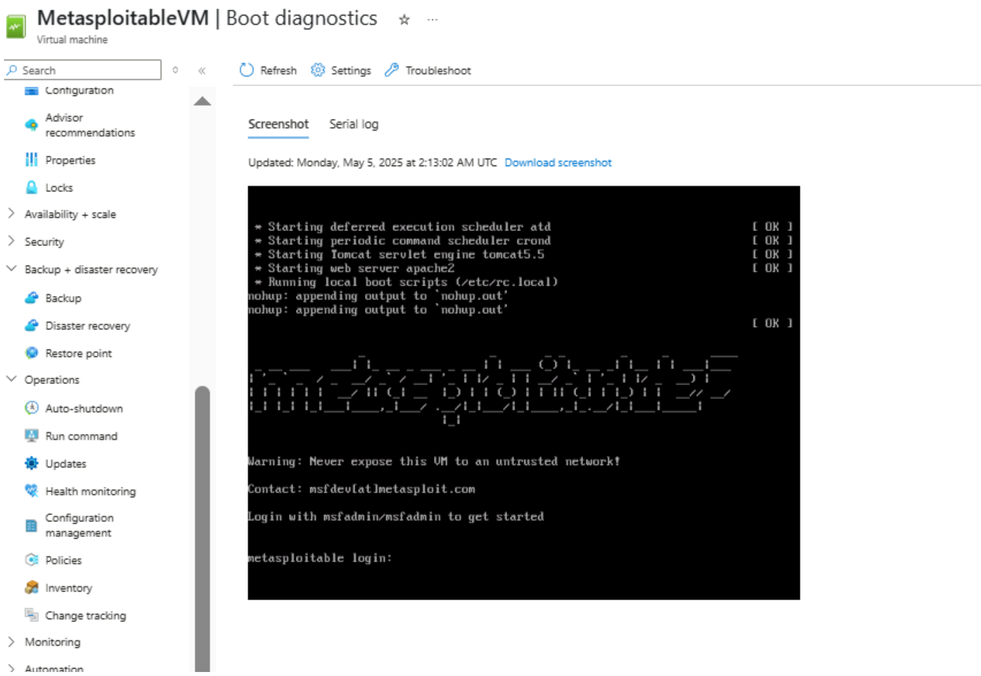


Disk Corruption During Upload: Despite successful local hash verification, VHD files showed corruption post-upload. This was attributed to Azure Storage Explorer's handling of large files and timeout issues, particularly with the Windows Server 2022 VHD exceeding 30GB.

Boot Configuration Incompatibility: The converted VHDs maintained VMware BIOS/UEFI settings, which are incompatible with Azure's Generation 1 and Generation 2 VM boot requirements and partition schemes.

Upload and Configuration Challenges

Storage Account Testing: Multiple approaches were tested using both page blobs and managed disks. Page blobs required specific 512-byte sector alignment that the conversion tool didn't consistently maintain. Zone redundancy was tested across different availability zones using Standard HDD and Premium SSD storage tiers to rule out disk performance issues.

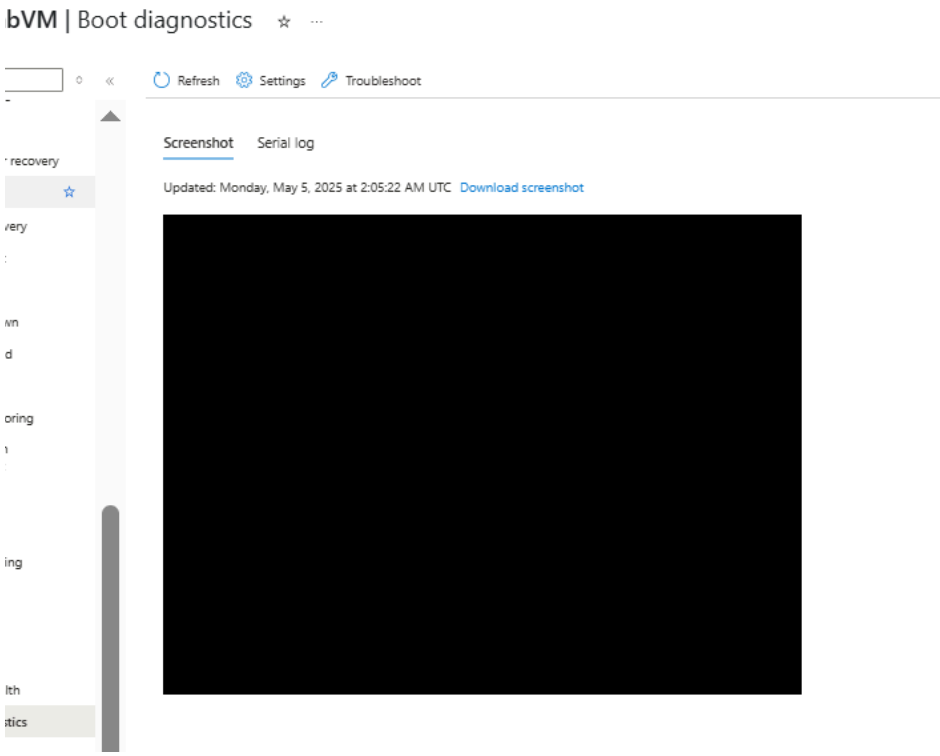


Azure Migrate Exploration: Attempted to use Azure Migrate for automated driver injection and boot configuration, but this required administrative access to configure the Azure Migrate appliance, necessitating IT department coordination.

Troubleshooting Methodology

Systematic troubleshooting included:

1. **Boot Diagnostics Analysis**: Used Azure serial console and boot diagnostic screenshots to identify exact failure points
2. **VHD Integrity Verification**: Performed checksum validation and partition table inspection
3. **Configuration Testing**: Tried various VM sizes, disk controllers (SCSI vs. IDE), and network configurations
4. **Snapshot Strategy**: Created incremental snapshots for rapid rollback during testing



Lessons Learned and Knowledge Gained

Technical Expertise Development

This migration provided hands-on experience with cloud infrastructure beyond classroom theory:

- Hypervisor Architecture: Deep understanding of differences between VMware and Azure's Hyper-V infrastructure, including hardware abstraction, virtual hardware presentation, and boot processes
- Azure Storage Technologies: Practical experience with Blob Storage, managed disks, page blobs, storage tiers, and proper VHD formatting requirements
- Boot Architecture: In-depth knowledge of Windows boot processes, MBR vs. GPT partitioning, UEFI vs. BIOS interfaces, and cloud hypervisor interactions

Process Insights

Tool Selection: Not all migration tools are equal. StarWind V2V, while functional for basic conversions, lacks enterprise features for production cloud migrations. Future efforts would benefit from Azure Migrate or enterprise-grade tools like Zerto.

Documentation Value: Maintaining detailed logs of configuration changes, error messages, and troubleshooting steps proved essential for pattern identification and avoiding repeated mistakes.

Professional Growth

This experience developed critical competencies:

- Problem-solving within Azure subscription limits and access constraints
- Resilience through iterative troubleshooting and strategy adaptation
- Real-world understanding of why cloud migrations require specialized expertise
- Appreciation for hybrid cloud complexity and phased migration approaches

The troubleshooting process became a practical master class in cloud infrastructure, storage management, and virtualization technology that will prove invaluable in cybersecurity and cloud operations.